

Optimizing Inventory and International Logistics for Manufacturing Operations

A Business Analytics & Supply Chain Analytics Portfolio Project

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Tools	Python, Microsoft Excel
Primary Dataset	Public Supply Chain Logistics Dataset

Executive Summary

Efficient inventory management and international logistics are fundamental to the success of modern manufacturing companies. Organizations must balance customer demand, inventory availability, production capacity, transportation cost, and supplier performance while remaining competitive in global markets. As supply chains become increasingly complex, companies are relying on Business Analytics to transform operational data into actionable insights that improve decision-making.

This portfolio project was inspired by a supply chain topic introduced during my internship at a manufacturing company with both production and sales operations. One discussion assigned to the internship team focused on inventory optimization and international logistics. Rather than using confidential company information, this project recreates a similar business scenario using a public supply chain dataset from Kaggle. The objective is to demonstrate analytical skills while protecting company confidentiality.

The project applies Business Analytics techniques including exploratory data analysis, inventory optimization concepts, logistics performance evaluation, and business visualization. Python and Microsoft Excel are applied throughout this project to analyze operational performance and develop recommendations for inventory planning, warehouse utilization, supplier management, and transportation efficiency.

Table 1. Project Overview

Table 1 provides a high-level overview of the project, including its business domain, analytical tools, datasets, and expected deliverables.

Category	Description
Industry	Manufacturing
Business Domain	Supply Chain & International Logistics
Primary Dataset	Public Supply Chain Logistics Dataset
Tools	Python, Microsoft Excel
Project Inspiration	Manufacturing Internship
Deliverables	Report, Python, README

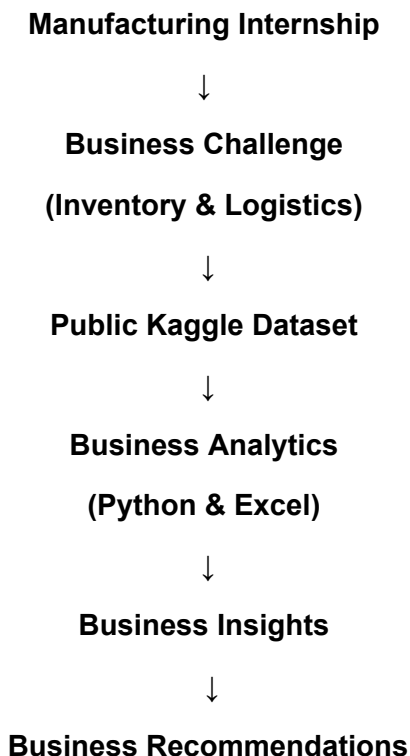
The table summarizes the overall scope of the project

1.Introduction

Manufacturing companies operate within highly interconnected supply chains where raw materials, suppliers, warehouses, transportation providers, production facilities, and customers all influence business performance. Even small inefficiencies in inventory planning or logistics coordination can significantly increase operating costs. Excess inventory ties up working capital and increases storage expenses, while insufficient inventory may interrupt production or delay customer orders.

International logistics creates additional challenges. Companies must manage transportation modes, customs procedures, delivery schedules, and supplier lead times while maintaining competitive service levels. Business Analytics provides managers with the ability to monitor operational performance, identify inefficiencies, and support data-driven decision-making instead of relying only on intuition.

This project investigates how publicly available supply chain data can be analyzed to support inventory optimization and international logistics decisions within a manufacturing environment. Although the analysis is based on a public dataset, the overall business scenario reflects practical challenges commonly encountered by manufacturing organizations.



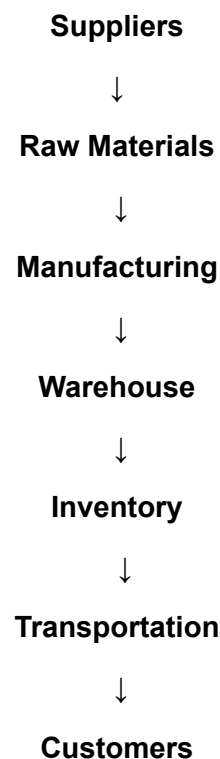
The above figure illustrates the overall development process of this portfolio project. The project originated from a real business topic encountered during an internship and was recreated using a public dataset to demonstrate Business Analytics skills without exposing confidential company information.

2. Manufacturing Industry Background

Manufacturing remains one of the most important sectors in the global economy. Manufacturing companies transform raw materials into finished goods through coordinated production processes supported by procurement, inventory management, warehousing, transportation, and customer fulfillment. Each stage of the supply chain contributes to operational performance and overall profitability.

Inventory management is particularly important because inventory represents both an operational necessity and a financial investment. Companies must determine how much inventory to purchase, when to replenish stock, and how to minimize carrying costs while maintaining service quality. International logistics also plays a significant role, especially for companies that source materials globally or export finished products. Transportation cost, shipping reliability, customs delays, and supplier performance all influence business outcomes.

Modern manufacturing companies increasingly adopt analytics technologies to improve forecasting, monitor key performance indicators, evaluate supplier reliability, and optimize inventory policies. Data-driven decision-making has become an essential competitive advantage for organizations seeking to improve efficiency while responding to changing customer demand.



The figure presents a simplified manufacturing supply chain. Each stage contributes to operational performance, making inventory planning and logistics management critical components of business success.

Table 2. Major Supply Chain Components

Table 2 summarizes the major components of a manufacturing supply chain and highlights the Business Analytics opportunities associated with each operational stage.

Component	Business Function	Analytics Opportunity
Suppliers	Provide raw materials	Supplier performance analysis
Manufacturing	Produce finished products	Production efficiency analysis
Warehouse	Store finished goods	Warehouse utilization
Inventory	Support customer demand	Safety Stock, Reorder Point
Transportation	Deliver products	Shipping cost optimization
Customers	Generate market demand	Demand forecasting

Table 2 highlights the key operational stages within a manufacturing supply chain and demonstrates where Business Analytics can provide decision support. Each component presents opportunities to improve efficiency through data analysis, helping organizations reduce costs, improve delivery performance, and better satisfy customer demand.

3. Project Background

During my internship at a manufacturing company with both production and sales operations, one of the business topics introduced to the internship team focused on improving inventory management and international logistics. The discussion highlighted how inventory planning, supplier coordination, warehouse operations, and transportation decisions affect operational efficiency and customer satisfaction.

That experience motivated me to explore these topics in greater depth. Because company information is confidential, this project does not use any internal data or operational records. Instead, it recreates a similar business scenario using a public supply chain logistics dataset from Kaggle. The dataset provides order, warehouse, freight, plant, and logistics data used throughout the analysis.

The purpose of this project is to demonstrate Business Analytics skills by applying practical supply chain concepts to realistic operational data while maintaining professional and ethical standards regarding company confidentiality.

4. Business Problems

Manufacturing companies commonly encounter several operational challenges. Excess inventory increases storage cost and ties up valuable working capital, while inventory shortages may interrupt production and reduce customer satisfaction. Long supplier lead times and transportation delays can further increase operational risk.

Warehouse utilization is another important issue. Poor inventory organization may reduce picking efficiency and increase fulfillment time. In addition, organizations must evaluate supplier performance, transportation methods, and logistics cost to maintain profitability in competitive markets.

This project focuses on understanding these operational challenges through data analytics. Rather than proposing theoretical solutions alone, the analysis aims to identify measurable opportunities for improvement using inventory metrics, logistics indicators, and supply chain performance data.

5. Project Objectives

The primary objective of this project is to demonstrate how Business Analytics techniques can improve inventory planning and international logistics within a manufacturing environment. More specifically, this project aims to evaluate inventory performance, examine logistics efficiency, identify operational bottlenecks, and provide practical recommendations using publicly available supply chain data.

The project also serves as a personal portfolio demonstrating analytical skills developed through academic study and internship experience. By integrating Python and Microsoft Excel, the project follows a complete analytics workflow from raw data preparation to business recommendations.

6. Dataset Description

Primary Dataset:

The project uses a public Supply Chain Logistics dataset stored in an Excel workbook. The workbook contains operational information including customer orders, warehouse costs, freight rates, plant information, transportation routes, and inventory-related records. This dataset provides sufficient information to evaluate inventory management and international logistics performance.

Table 3. Primary Dataset Overview

Table 3 summarizes the primary dataset used in this project and explains its role within the overall analytical framework.

Dataset	Purpose	Reason for Selection
Public Supply Chain Logistics Dataset	Primary Dataset	Contains order, warehouse, freight, plant, and logistics data used throughout the project.

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This particular data set was chosen due to its wide scope of operation, which covers customer orders, delivery performance, plant operations, warehouse costs and capacities, freight charges, and shipping routes.

Table 4. Key Variables Used in Analysis

Table 4 lists the key variables selected from the datasets and explains their business purpose throughout the analytical process.

Variable	Description	Business Purpose
Order Date	Date of customer order	Analyze order trends over time
Product ID	Unique product identifier	Analyze product demand and support ABC classification
Unit Quantity	Number of units ordered	Measure product and order volume
Carrier	Transportation provider	Evaluate carrier activity and performance
Plant Code	Manufacturing or distribution plant identifier	Compare plant-level operational activity
Service Level	Shipment service category	Evaluate logistics service patterns
Ship Late Day Count	Number of days a shipment was late	Measure delivery performance
Freight Rate	Transportation rate associated with shipments	Compare transportation costs
Warehouse Cost per Unit	Cost of warehouse operations per unit	Compare warehouse costs
Daily Capacity	Daily operational capacity of a plant	Evaluate capacity across plants

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The above-mentioned variables constitute the major operational indicators that would be employed to assess order patterns, product demand, deliveries, carriers, plant activities, warehousing expenses, capacity, and transportation performance. The variables form the basis for the inventory and logistics analysis carried out in this project.

7. Methodology

The project follows a structured Business Analytics framework consisting of seven stages.

Stage 1: Business Understanding

Identify the operational challenges associated with inventory management and international logistics.

Stage 2: Data Collection

Obtain publicly available supply chain datasets from Kaggle while protecting confidential company information.

Stage 3: Data Preparation

Perform data cleaning, validation, and data type conversion to improve data quality and analytical consistency.

Stage 4: Exploratory Data Analysis

Analyze order patterns, product demand, delivery performance, carrier performance, plant operations, warehouse costs and capacities, and freight rates.

Stage 5: Inventory Optimization

Apply EOQ, Safety Stock, Reorder Point (ROP), and ABC Classification to demonstrate inventory planning concepts .

Stage 6: Visualization

Develop business visualizations using Python.

Stage 7: Business Recommendations

Interpret analytical findings and provide practical recommendations for operational improvement.

8. Data Cleaning

Data quality is necessary for accurate business analytics. Preprocessing of the datasets occurred before any of the analyses to ensure that high data quality standards were achieved. Standardization of column names, deletion of full rows and columns without any data, elimination of duplicates, and elimination of extra white space in text variables were some of the actions taken. Some key date and numerical variables were also converted to the appropriate data types.

Table 5. Data Cleaning Summary

Table 5 summarizes the major data cleaning procedures performed before analysis to improve data quality and analytical reliability.

Cleaning Task	Purpose
Remove Empty Rows and Columns	Improve data completeness
Remove Duplicate Records	Prevent duplicate analysis
Standardize Column Names	Ensure analytical consistency
Clean Text Fields	Ensure consistent text values
Convert Data Types	Ensure accurate analysis and calculations

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Data cleaning is one of the most important stages in Business Analytics because poor-quality data can lead to misleading conclusions. A well-prepared dataset provides a reliable foundation for inventory optimization and logistics analysis.

9. Exploratory Data Analysis (EDA)

Exploratory Data Analysis provides the foundation for understanding operational performance before optimization techniques are applied.

The exploratory analysis focuses on monthly order volume, product demand, delivery performance, carrier performance, plant performance, warehouse costs and capacities, freight rates, transportation routes, and plant network characteristics.

Table 6. Key Exploratory Analysis

Table 6 summarizes the key exploratory and operational analyses conducted to evaluate product demand, delivery performance, plant operations, warehouse efficiency, and logistics performance.

Analysis	Business Purpose
Monthly Order Analysis	Identify order volume trends over time
Product Analysis	Identify high-volume products
ABC Inventory Classification	Prioritize products based on unit volume
Delivery Performance Analysis	Evaluate shipment reliability and late deliveries
Carrier Performance Analysis	Compare carrier activity and delivery performance
Plant Performance Analysis	Evaluate operational activity across plants
Warehouse Cost Analysis	Compare warehouse cost per unit
Warehouse Capacity Analysis	Evaluate daily capacity across plants
Freight Rate Analysis	Compare transportation costs across shipping modes
Freight Route Analysis	Evaluate cost differences across transportation routes
Plant Network Analysis	Evaluate product, customer, and port coverage across plants

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Analysis of these studies offers a detailed understanding of product demand, service dependability, carrier activities, production process, cost and capacity of warehouses, and transport activities. These results support the recommendations on inventory optimization and logistics management discussed in subsequent sections.

10. Inventory Optimization

Inventory optimization is one of the primary objectives of this project because inventory directly affects operating cost, production continuity, and customer service. The analysis focuses on identifying opportunities to improve inventory planning through several widely adopted supply chain concepts.

Economic Order Quantity (EOQ) is introduced as one of the inventory optimization methods evaluated in this project. Although the selected public datasets do not contain every financial variable required for a complete EOQ implementation, example calculations will demonstrate how the method supports purchasing decisions.

Safety Stock analysis evaluates the importance of maintaining additional inventory to reduce stockout risk during demand uncertainty or supplier delays. Products with highly variable demand or long lead times generally require larger safety stock levels.

Reorder Point (ROP) analysis combines average demand with supplier lead time to estimate when replenishment should begin. This approach reduces the probability of inventory shortages while avoiding unnecessary inventory accumulation. Since lead times are unavailable in the dataset, transportation time (TPT) is taken to be the proxy for lead time in the Safety Stock and Reorder Point examples. A 95% service level is considered.

ABC Inventory Classification System classifies inventory based on unit volume and activity. Products that fall under the category of A have high unit volumes, thus requiring more focus from management. Category A items receive the highest management attention because they contribute the greatest business impact. Category B items require periodic review, while Category C items can often be managed with simplified inventory policies.

Together these methods demonstrate how Business Analytics can support inventory planning and improve operational efficiency.

Table 7. Inventory Performance Indicators

Table 7 presents the primary inventory performance indicators used to evaluate inventory efficiency and support operational decision-making.

KPI	Business Purpose	Expected Benefit
Safety Stock	Prevent stock shortages	Improve customer service
Reorder Point (ROP)	Determine replenishment timing	Reduce stockout risk
EOQ	Optimize purchasing quantity	Lower inventory costs
Warehouse Utilization	Measure storage efficiency	Improve warehouse capacity
Order Fulfillment Rate	Evaluate customer service	Increase customer satisfaction

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The inventory and logistics KPIs presented in Table 7 provide measurable indicators for evaluating inventory efficiency, replenishment planning, warehouse utilization, and order fulfillment performance. Monitoring these metrics regularly enables organizations to balance inventory costs with customer demand while improving overall supply chain efficiency.

11. International Logistics Analysis

International logistics influences both customer satisfaction and total operating cost. Manufacturing companies frequently choose between multiple transportation options based on delivery speed, shipping cost, and service requirements.

This study analyzes the logistics performance through the analysis of delivery results, carrier performance, freight pricing, routing of transportations, plant performance, warehouse costing, and warehouse capacity. The late deliveries are evaluated to find out any possible problems with service, while the analysis of carriers and freight helps understand the performance of transportations and costs. Analysis of plants and warehouses is conducted for the evaluation of performance and capacity.



The figure illustrates the major stages of international logistics. Each stage influences delivery performance, transportation costs, and customer satisfaction, making logistics optimization a key business objective.

12. Business Recommendations

Based on the analytical results, the following recommendations are proposed.

1. Adopt inventory policies based on data rather than intuition.
2. Review EOQ assumptions periodically and update ordering and holding cost estimates when more detailed operational data become available.
3. Use ABC Classification to prioritize high-volume products for closer inventory monitoring.
4. Monitor carrier and delivery performance using shipment volume and late-delivery indicators.
5. Evaluate transportation mode selection according to cost and service requirements.
6. Improve warehouse visibility using dashboard reporting and KPI monitoring.
7. Standardize inventory review processes across warehouses.
8. Continue using Python for automated reporting and consider adding SQL for future data management and analysis.
9. Develop interactive Tableau dashboards for managers.
10. Expand future analysis with demand forecasting models and ERP integration.

Table 8. Recommendation Summary

Table 8 summarizes the recommended business actions based on the inventory and logistics analyses conducted in this project.

Recommendation	Expected Business Benefit
Inventory Optimization	Improve inventory planning using EOQ, Safety Stock, Reorder Point, and ABC Classification
Warehouse Monitoring	Improve visibility of warehouse capacity and operating costs
Carrier Performance Monitoring	Improve delivery reliability by tracking carrier and late-shipment performance
Transportation Optimization	Reduce logistics costs by comparing freight rates and transportation routes
Automated Reporting	Improve reporting efficiency through Python-based analytics
Tableau Dashboard	Provide interactive visualization of key supply chain KPIs
Predictive Analytics	Support future demand forecasting and inventory planning

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These recommendations focus on improving inventory planning, warehouse operations, delivery performance, transportation efficiency, and data-driven decision-making.

13. Future Improvements

This project demonstrates how Business Analytics techniques can be applied to inventory optimization and international logistics using a publicly available dataset. However, several opportunities remain for future development.

Future improvements include integrating real-time ERP data, incorporating supplier performance metrics, developing machine learning forecasting models, and expanding the analysis to include procurement optimization and production scheduling. Interactive Tableau dashboards can also be developed to visualize additional KPIs and support management reporting.

As analytical tools continue to evolve, organizations can further improve decision-making by combining Business Intelligence, predictive analytics, and artificial intelligence technologies.

Table 9. Future Enhancement Plan

Table 9 outlines several future enhancement opportunities that could further strengthen inventory optimization and supply chain analytics capabilities.

Future Enhancement	Business Value
Machine Learning Forecasting	Improve demand prediction
ERP Data Integration	Enable real-time analytics
Supplier Performance Dashboard	Improve supplier management
Automated Reporting	Increase reporting efficiency
AI-based Inventory Optimization	Support smarter decisions

14. Conclusion

This project demonstrates how Business Analytics techniques can be applied to inventory optimization and international logistics within a manufacturing environment. Inspired by a real internship discussion but developed entirely with a public dataset, the project illustrates an ethical approach to portfolio development while protecting company confidentiality.

The final portfolio combines business understanding, analytical methodology, and practical supply chain concepts to simulate challenges commonly encountered by manufacturing organizations. The final Python project proves that operational supply chain data can be converted into useful business information. Possible improvements could involve SQL coding, Tableau dashboards, demand forecasting, and more quantitative analyses. This could help improve the overall project for submission on GitHub and LinkedIn and even for applying to graduate school and future jobs.

15. References

- Chopra, S., & Meindl, P. (2022). *Supply chain management: Strategy, planning, and operation* (8th ed.). Pearson.
<https://www.kaggle.com/datasets/anisseezzebdi/supply-chain-logistics-problem>